

Comparative Study of Rule based, classical supervised machine learning and transformer based sentiment analysis.

Fadi Awawdeh

Abstract

Sentiment analysis is a core natural language processing task with widespread industrial adoption. Despite recent dominance of transformer based architectures, simpler models remain attractive due to efficiency and interpretability advantages. This paper presents a controlled comparative study of rule-based, classical machine learning, deep learning and transformer based sentiment classifiers across three benchmark datasets: IMDB Movie Reviews, Stanford Sentiment Treebank (SST-5), and Amazon Customer Reviews. VADER, Naive Bayes, Logistic Regression, convolutional neural networks (CNNs), and RoBERTa were evaluated using consistent metrics and experimental conditions. Results show that optimized classical models achieve maximum performance on large scale binary sentiment tasks, matching or exceeding transformer performance with substantially lower computational cost. Conversely, fine grained sentiment classification benefits from contextual transformer representations. We demonstrate that dataset size and sentiment granularity are the primary drivers of model effectiveness, through quantitative and qualitative error analysis, highlighting the importance of principled model selection over architectural complexity.

1. Introduction

Sentiment analysis aims to identify the polarity or emotional tone expressed in text and remains a foundational task in natural language processing (NLP). Applications include opinion mining, customer feedback analysis and recommender systems. While transformer based models have become the accepted practice for many NLP tasks, their high computational cost and limited interpretability raise practical concerns for deployment.

A fundamental trade off exists between accuracy, efficiency and interpretability. Rule based and classical machine learning approaches offer transparency and fast inference but may fail to capture contextual nuance. Meanwhile deep learning and transformer based models capture complex linguistic patterns but require substantial data and computational resources.

Research Question.

How do rule based, classical supervised machine learning and transformer based models differ in accuracy and efficiency for sentiment classification across multiple datasets?

Contributions.

This work provides a unified comparison of five model families across three datasets, evaluates the impact of hyperparameter optimization and performs deep error analysis to identify systematic model failure modes.

2. Related Work

Early sentiment analysis research relied on lexicon based methods and linear classifiers (Pang and Lee, 2008). Classical approaches such as Naive Bayes and Logistic Regression with bag of words or TF-IDF features remain strong baselines due to their simplicity and robustness. Neural architectures, particularly convolutional neural networks, demonstrated improvements by capturing local n-gram patterns when sufficient labeled data is available (Kim, 2014).

Transformer based models, including BERT and RoBERTa, leverage large scale pretraining and contextualized representations, achieving what so called state of the art results on sentiment benchmarks (Devlin et al., 2019; Liu et al., 2019). However, recent studies suggest that classical models remain competitive on large-scale binary sentiment tasks, especially when computational

efficiency is prioritized. This work differs by directly comparing these paradigms under consistent experimental settings across datasets of varying scale and complexity.

3. Methodology

3.1 Datasets

Three English language datasets are used:

- **IMDB Movie Reviews:** 50,000 balanced binary-labeled reviews.
 - **Stanford Sentiment Treebank (SST-5):** ~11,000 sentences labeled into five sentiment classes.
 - **Amazon Customer Reviews:** Large scale binary sentiment dataset of reviews. Standard train validation test splits are used.
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3.2 Preprocessing

For classical models, text is lowercased, tokenized, lemmatized, and vectorized using TF-IDF. Neural models use fixed length sequences with padding; CNNs employ pre trained GloVe embeddings, while RoBERTa uses WordPiece tokenization.

3.3 Models and Training

VADER: Rule based sentiment scoring without training.

Naive Bayes: Multinomial NB and ComplementNB variants.

Logistic Regression: L2 regularized classifier optimized via grid search.

CNN: One dimensional convolutional architecture.

RoBERTa: Evaluated in zero shot and fine tuned settings using learning rates.

Models are evaluated using accuracy, precision, recall and F1-score.

4. Results and Analysis

4.1 Overall Performance

On IMDB and Amazon datasets, optimized Logistic Regression achieves 96–98% accuracy, matching or exceeding RoBERTa performance with significantly lower computational cost. On SST, transformer based models outperform classical approaches, achieving approximately 90% accuracy due to superior contextual modeling.

4.2 Quantitative Error Analysis

Confusion Matrices.

SST confusion matrices show frequent misclassification between adjacent sentiment classes (e.g., neutral vs. slightly positive). Classical models disproportionately collapse neutral sentiment into polar extremes.

Per-Class Metrics.

ComplementNB improves recall for minority sentiment classes on SST, while Logistic Regression exhibits reduced recall for neutral sentiment. CNNs demonstrate high recall but lower precision particularly for negative classes.

Performance on Data Slices.

Short texts (<10 tokens): VADER performs competitively.

Long reviews (>200 tokens): VADER and CNN performance degrades due to sentiment drift.

Negation heavy sentences: RoBERTa consistently outperforms classical models.

4.3 Qualitative Error Analysis

Manual Error Categorization.

Inspection of 30 misclassified examples reveals recurring error sources: sarcasm, mixed sentiment, domain specific language and annotation noise.

Case Studies.

“I expected this movie to be terrible, but it wasn’t.”

Logistic Regression predicts negative due to lexical cues; RoBERTa correctly predicts positive.

“The product works, but the customer service was awful.”

Binary classifiers struggle with mixed sentiment signals.

“Not bad at all.”

VADER misclassifies due to limitations in negation handling.

5. Discussion and Future Work

Results indicate that dataset size and sentiment granularity dominate model effectiveness. Classical models scale efficiently with large datasets, while transformers provide clear benefits for nuanced sentiment tasks. Limitations include English only evaluation and absence of aspect level sentiment analysis. Future work should explore cross lingual settings, aspect based sentiment classification and hybrid classical transformer approaches.

6. Conclusion

This study demonstrates that architectural complexity is not universally necessary for strong sentiment analysis performance. Optimized Logistic Regression matches transformer based models on large scale binary tasks while remaining efficient and interpretable. Transformer models are most beneficial for fine grained or context dependent sentiment classification. These findings emphasize the importance of empirical justification when selecting model complexity for real world NLP applications.

References

- Pang, B., & Lee, L. (2008). *Opinion Mining and Sentiment Analysis*.
- Kim, Y. (2014). *Convolutional Neural Networks for Sentence Classification*.
- Devlin, J., et al. (2019). *BERT: Pre-training of Deep Bidirectional Transformers*.
- Liu, Y., et al. (2019). *RoBERTa: A Robustly Optimized BERT Pretraining Approach*.